D 92910–B

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Name.....

Reg. No.....

THIRD SEMESTER B.A./B.Sc. (CBCSS—UG) DEGREE EXAMINATION NOVEMBER 2020

Common Course

A11—BASIC NUMERICAL SKILLS

(2019 Admissions)

Time : Two Hours and a Half

Maximum : 80 Marks

Section A (Short Answers)

Answer at least **ten** *questions.*
Each question carries 3 *marks.*
All questions can be attended.
Overall Ceiling 30.

1. The sum of three numbers in AP is – 3 and their product is 8. Find the numbers.
2. What is an index number ?
3. Find the power set of A = {*a*, *b*, *c*}.
4. Find the value of the determinant $\begin{vmatrix} 2 & 0 & 1 \\ 0 & 1 & 1 \\ 2 & 1 & 2 \end{vmatrix}$.
5. Differentiate between discrete and continuous frequency distributions ?
6. Explain Kurtosis.
7. What do you understand by classificationof data ?
8. Find the mean of the following data :

   4, 45, 60, 20, 83, 19, 26, 11, 27, 12, 52.
9. If the 8th term of an AP is zero, prove that its 28th term is double the 18th term.
10. What is analysis of time series
11. Solve $x^2 - 5x + 4 = 0$ by using quadratic formula.
12. Differentiate Geometric and Harmonic Mean.

13. What is a pie diagram ?

14. Explain Skewness.

15. Represent the following frequency table by histogram.

Marks:	:	10 – 15	15 – 20	20 – 25	25 – 30	30 – 35
Number of students	:	20	20	30	20	50

(10 × 3 = 30 marks)

Section B (Paragraph)

Answer at least **five** *questions.*
Each question carries 6 *marks.*
All questions can be attended.
Overall Ceiling 30.

16. Find the sum of first 22 terms of the sequence 5, 10, 15, 20, ————.

17. Find the central tendencies for the given series :

3, 9, 3, 5, 12, 10, 18, 4, 7, 19, 21.

18. If A = {1, 2, 3} and B = {a, b}, find A × B and B × A. Are they equal ?

19. Find AB, Where $A = \begin{bmatrix} 1 & 2 \\ 2 & 1 \end{bmatrix}$ and $B = \begin{bmatrix} 2 & 1 \\ 1 & 1 \end{bmatrix}$.

20. Find the adjoint of the matrix $\begin{bmatrix} 1 & 1 & 2 \\ 0 & 2 & 3 \\ 0 & 0 & 1 \end{bmatrix}$.

21. Give three yearly moving averages for the following series :

Year	:	2005	2006	2007	2008	2009	2010	2011	2012	2013	2014
Production (lakh tons)	:	17.2	17.3	17.7	18.9	19.2	19.3	18.1	20.2	25.3	24.9

22. Find the sum of first 10 terms of GP, whose 3rd term is 12 and 8th term is 384.

23. What are the different aspects to be considered in planning a statistical enquiry ?

(5 × 6 = 30 marks)

Section C (Essay)

Answer any **two** *question.*
Each question carries 10 *marks.*

24. Explain :

    (a) Arithmetic, Geometric and Harmonic mean.

    (b) Scope of statistics.

25. Find the quartile deviation for the following data :

Marks		*Frequency*
0 – 5	...	4
5 – 10	...	5
10 – 15	...	6
15 – 20	...	10
20 – 25	...	11
25 – 30	...	9
30 – 35	...	4
35 – 40	...	1

26. Find the sum of n terms of the series $5 + 55 + 555 + 5555 + \ldots$

27. Find the inverse of the matrix $\begin{bmatrix} 0 & 1 & 2 \\ 1 & 2 & 3 \\ 3 & 1 & 1 \end{bmatrix}$.

(2 × 10 = 20 marks)